OptiVest AI - System Architecture

Market Data API
Yahoo Finance (yfinance)
Daily OHLCV, 106 Assets

User Portfolio
Holdings, Cash Balance
Risk Preferences

Asset Registry
5-Tier Universe
106 Active Assets

Feature Engineering Pipeline

35 Features: Price Returns, SMA/EMA, RSI, MACD, Bollinger Bands, Volatility, Volume, Calendar

Risk Tolerance
1-5 Scale

Market Regime
Bull, Bear
Sideways

Multi-Model Prediction Engine
LSTM, GRU, XGBoost, LightGBM
Random Forest, Extra Trees, Linear Regression
Multi-Horizon: 1D / 5D / 10D / 20D / 30D / 90D

Portfolio Intelligence
Covariance Matrix
Sharpe, Sortino, Max Drawdown
Beta, Sector Concentration

STAGE 1

Dynamic Risk-Adaptive Multi-Factor Scoring -> Top-N Candidate Pool

Candidate Discovery Engine (CDE) - Phase 18

- **Expected Return (ML)** *w=0.25*
- **Volatility (Risk)** *w=0.25*
- **Momentum (Trend)** *w=0.15*
- **Diversification (Correlation)** *w=0.20*
- **Fundamental Quality** *w=0.15*

STAGE 2

$$\max_{\Delta \mathbf{w}} \mathbf{w}_{new}^T \hat{\mathbf{r}} - \frac{\lambda}{2} \mathbf{w}_{new}^T \Sigma \mathbf{w}_{new} - \gamma \sum |\Delta w_i|$$

Adaptive Portfolio Decision Engine (APDE) - Phase 6

- **Turnover Cap** ||dw|| <= C_realloc
- **No Short-Selling** w_i >= 0
- **Budget Constraint** Sum(w_i) = 1
- **Sector Limits** Sum(w_sector) <= S_max
- **SLSQP Solver** scipy.optimize

STAGE 3

4-Dimensional Explainability Framework

Explainable AI Engine (XAI) - SHAP Attribution

- **TreeSHAP Feature Attribution**
- **Why This Stock?**
- **Why Not Another?**
- **Why Now? (Regime)**
- **Why This %? (Constraints)**

Decision Cards + Paper Trading Execution

Scored Recommendations (0-100), User Governance (Accept/Reject/Modify)
Simulated Execution, Performance Analytics, Research Reports